

Bandwidth Control Approaches

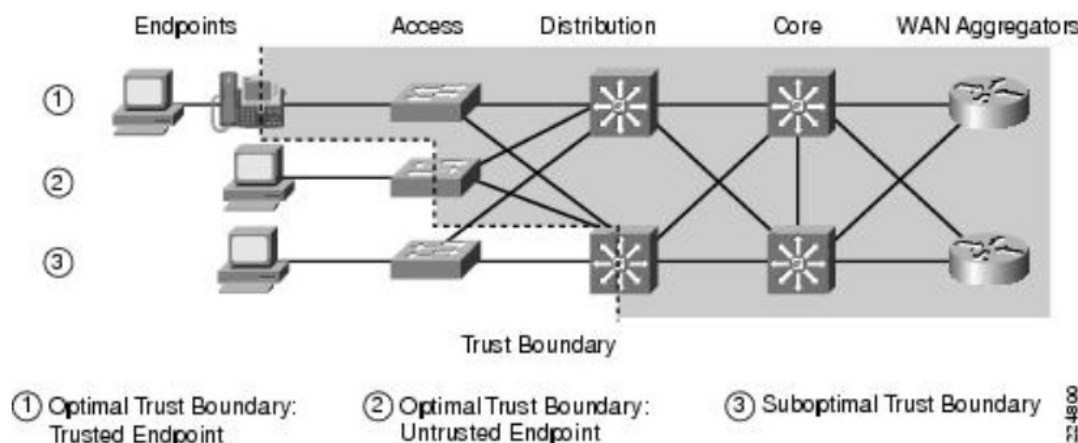
Cisco versus Mikrotik on Multitenancy

Abril de 2016

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Cisco - QoS Enterprise Design

Figure 2-3 Establishing Trust Boundaries



Acc. → Policy ↔ Agg. → Shaping

So where is QoS required in campus?

Access switches require the following QoS policies:

- Appropriate (endpoint-dependant) trust policies, and/or classification and marking policies
- Policing and markdown policies
- Queuing policies.

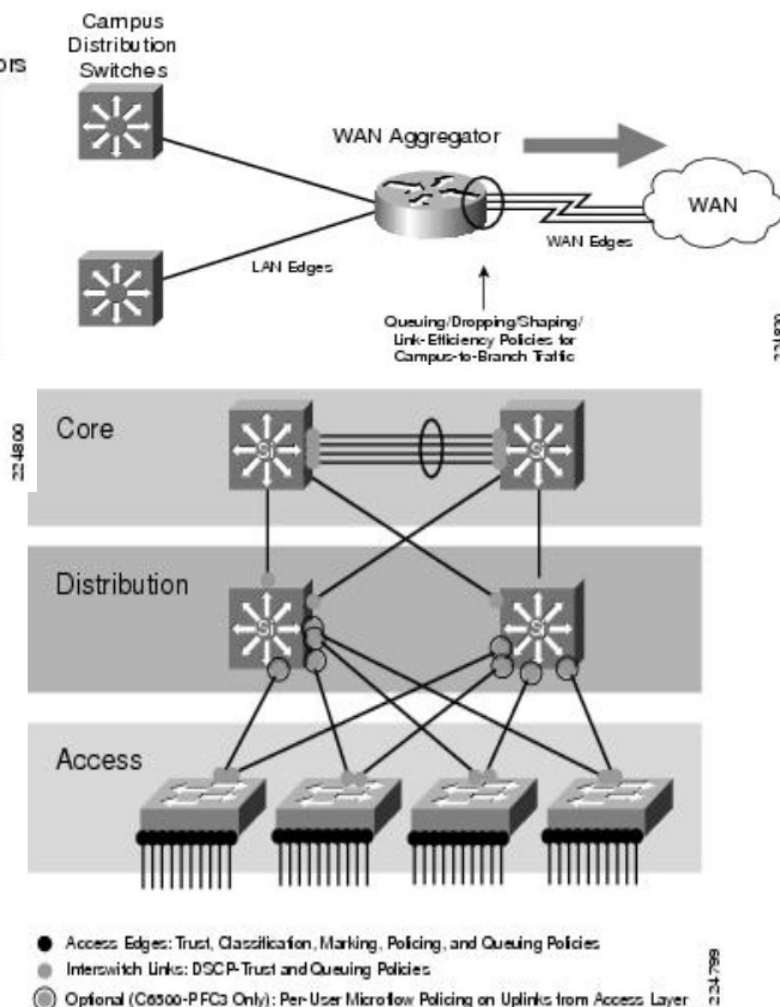
Distribution and core switches require the following QoS policies:

- DSCP trust policies
- Queuing policies
- Optional per-user microflow policing policies (only on supported platforms)

These recommendations are summarized in [Figure 2-2](#).

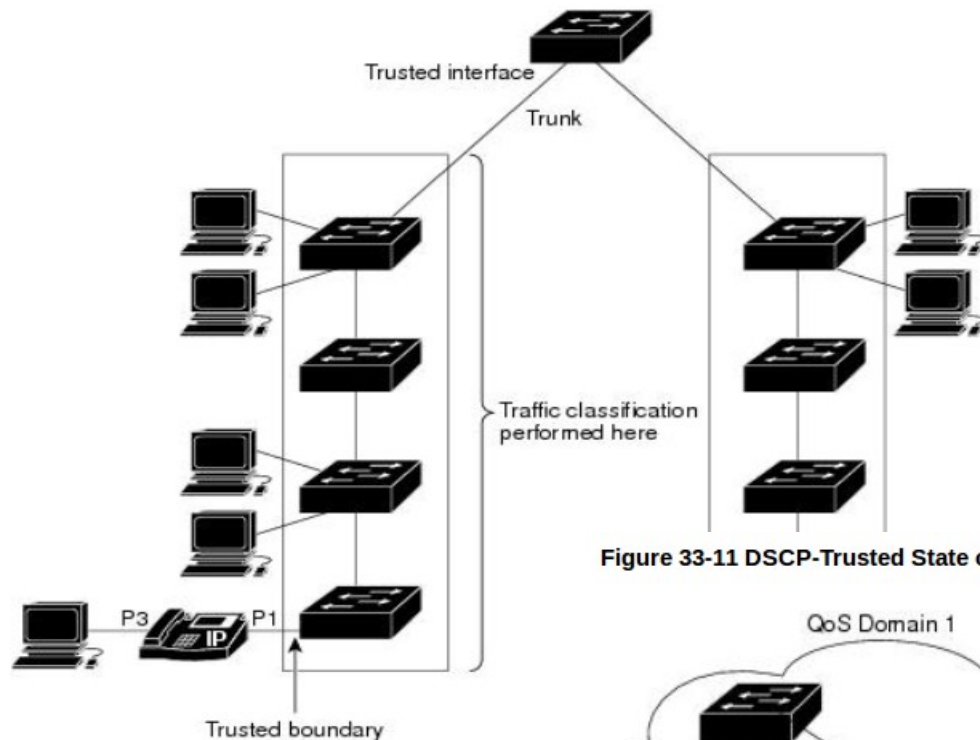
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http://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/WAN_and_MAN/QoS_SRND/QoS-SRND-Book/WANQoS.html

Figure 3-1 Where is QoS Needed over the WAN?



Cisco – QoS Multi-Tenancy

Figure 33-10 Port Trusted States within the QoS Domain



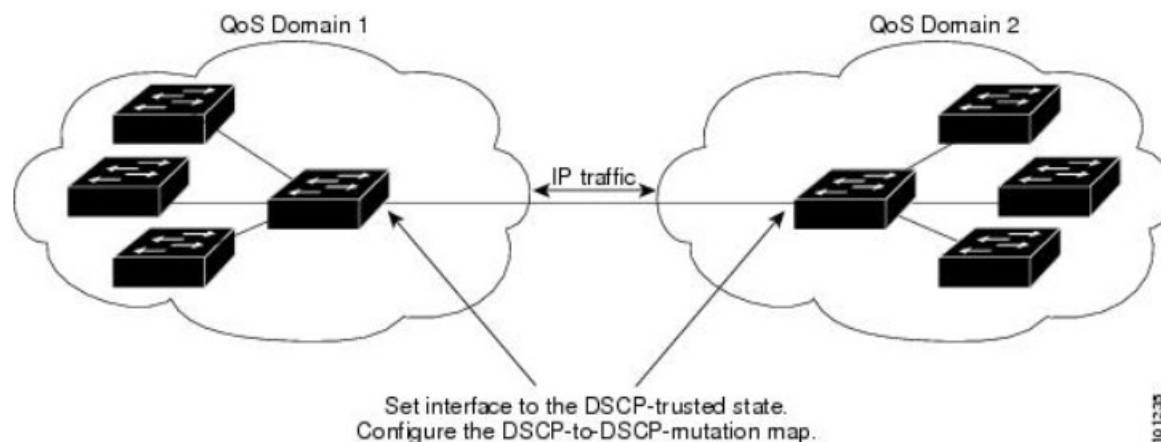
Bandwidth Provisioning

Real-Time (i.e. Default → VoIP)

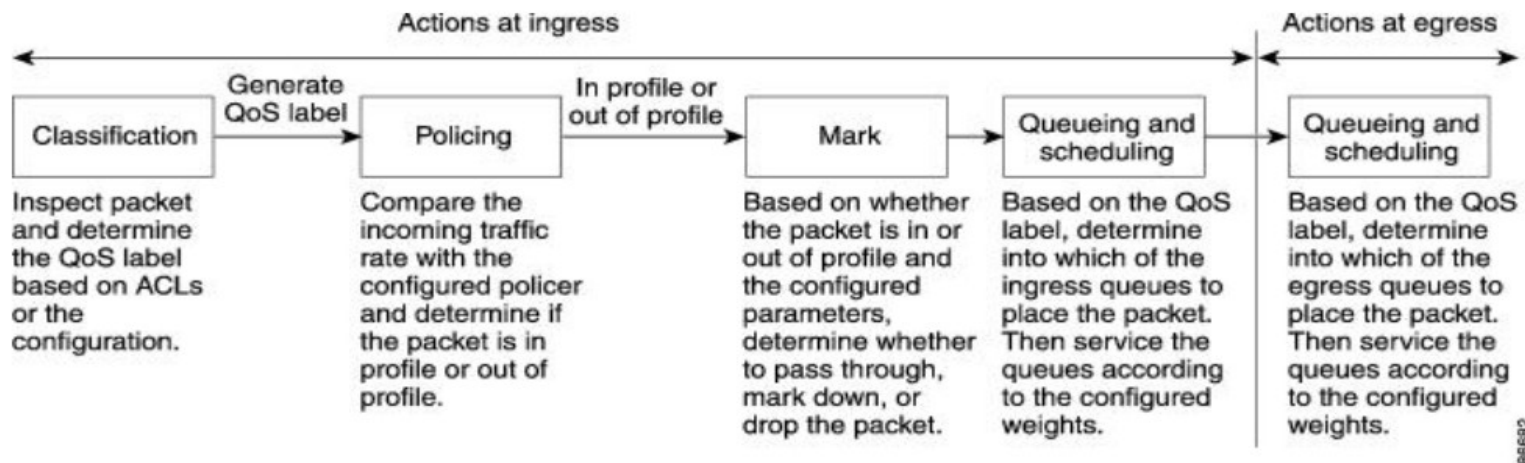
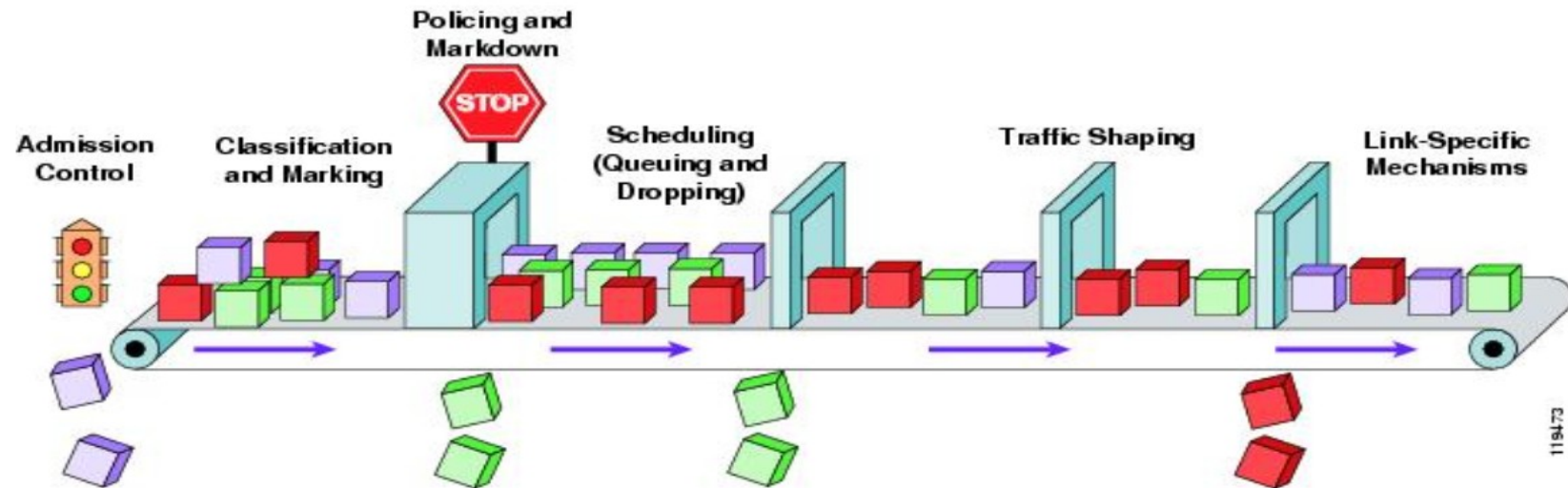
Best-Effort (i.e. Default → Data)

What about Multi-tenancy?

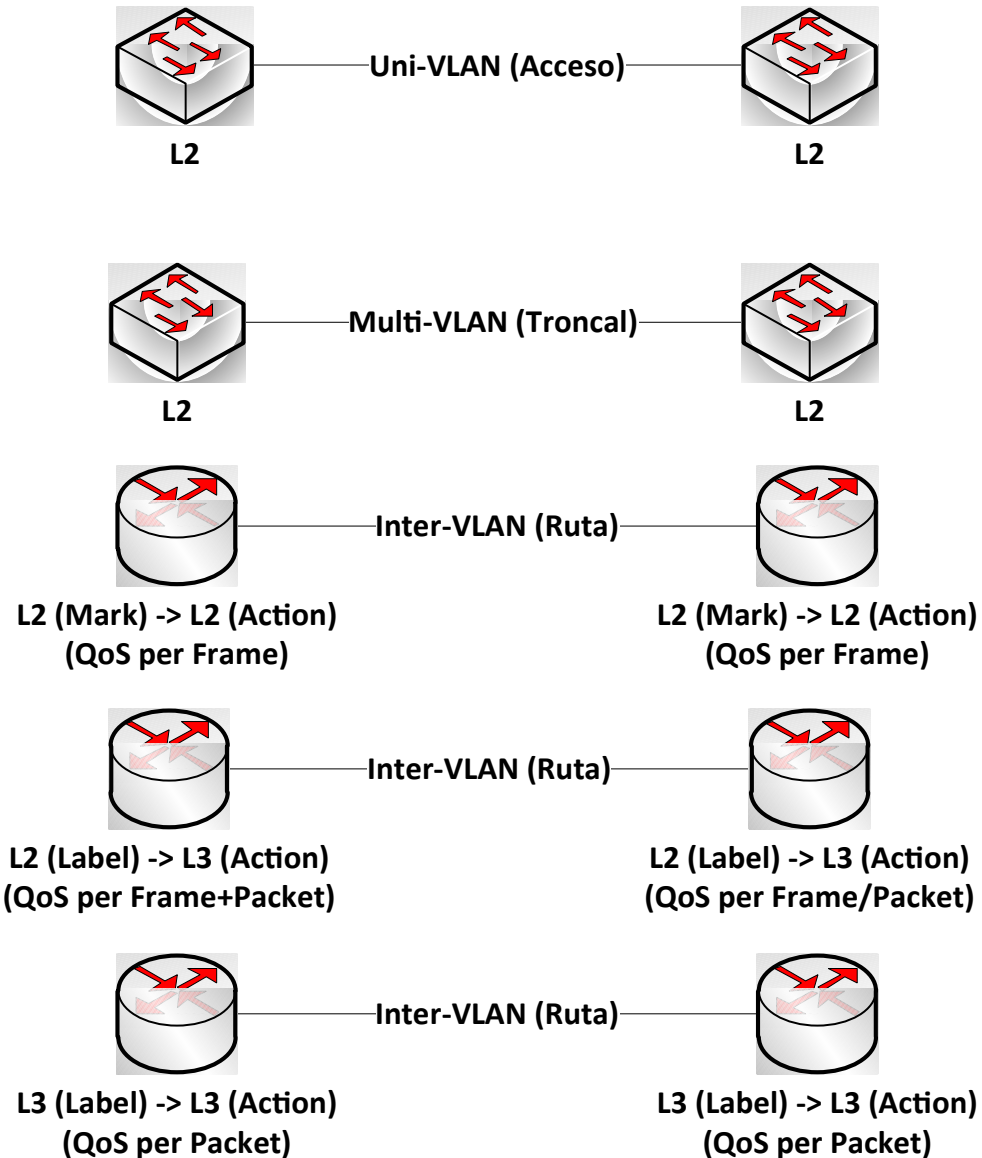
Figure 33-11 DSCP-Trusted State on a Port Bordering Another QoS Domain



Quality of Service (QoS)



QoS – On What?



Bridge-VLANs
(L2/Ethernet->Frames)



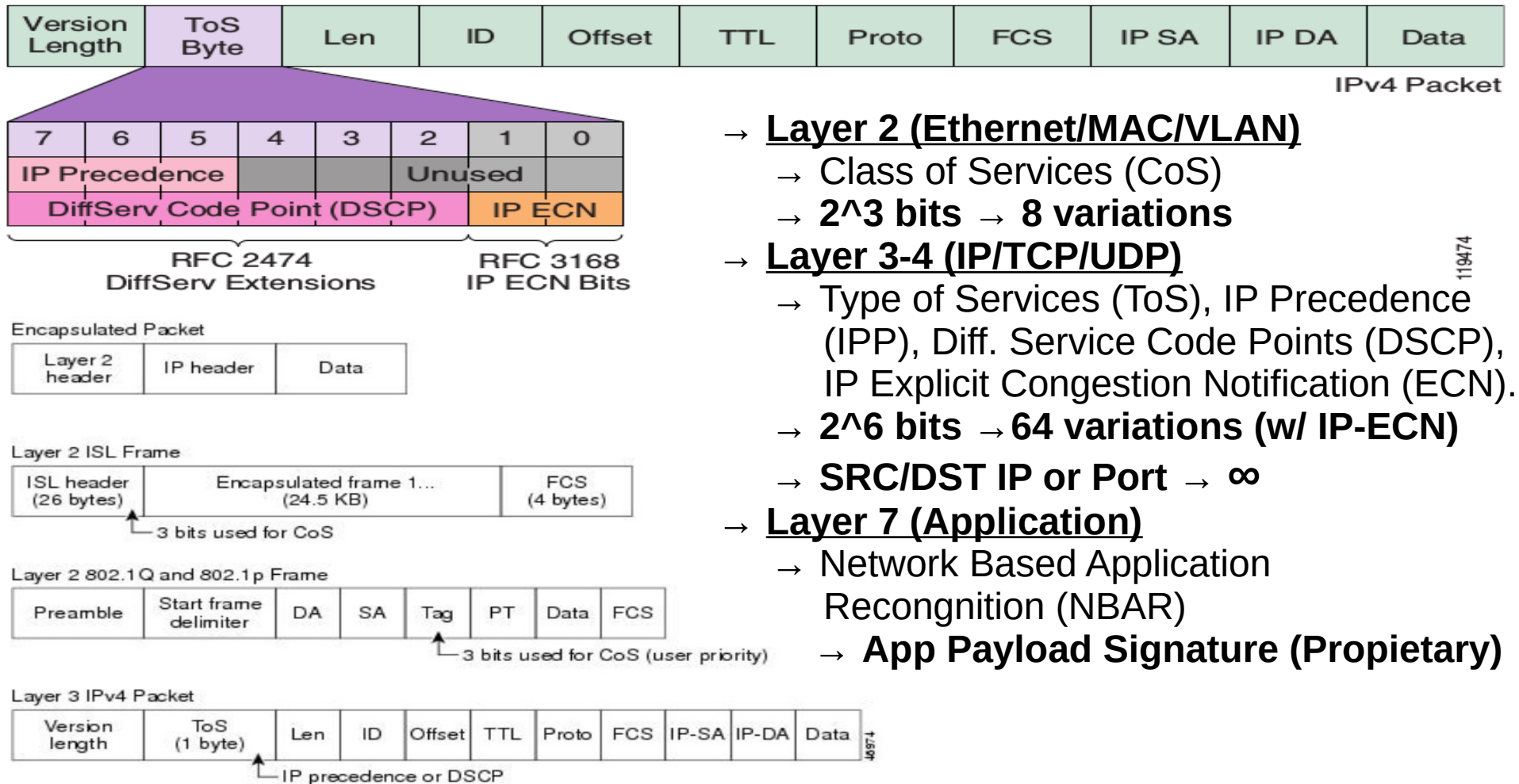
**Policy/
Shape**



Routed-VLANs
(L3/IP->Packets)

QoS - Labels for Classification and Marking

Figure 1-2 The IP ToS Byte (DSCP and IP ECN)



- **Layer 2 (Ethernet/MAC/VLAN)**
 - Class of Services (CoS)
 - **2³ bits → 8 variations**
- **Layer 3-4 (IP/TCP/UDP)**
 - Type of Services (ToS), IP Precedence (IPP), Diff. Service Code Points (DSCP), IP Explicit Congestion Notification (ECN).
 - **2⁶ bits → 64 variations (w/ IP-ECN)**
 - **SRC/DST IP or Port → ∞**
- **Layer 7 (Application)**
 - Network Based Application Recognition (NBAR)
 - **App Payload Signature (Proprietary)**

Cisco - QoS HW Escalability

Figure 33-5 Ingress and Egress Queue Location

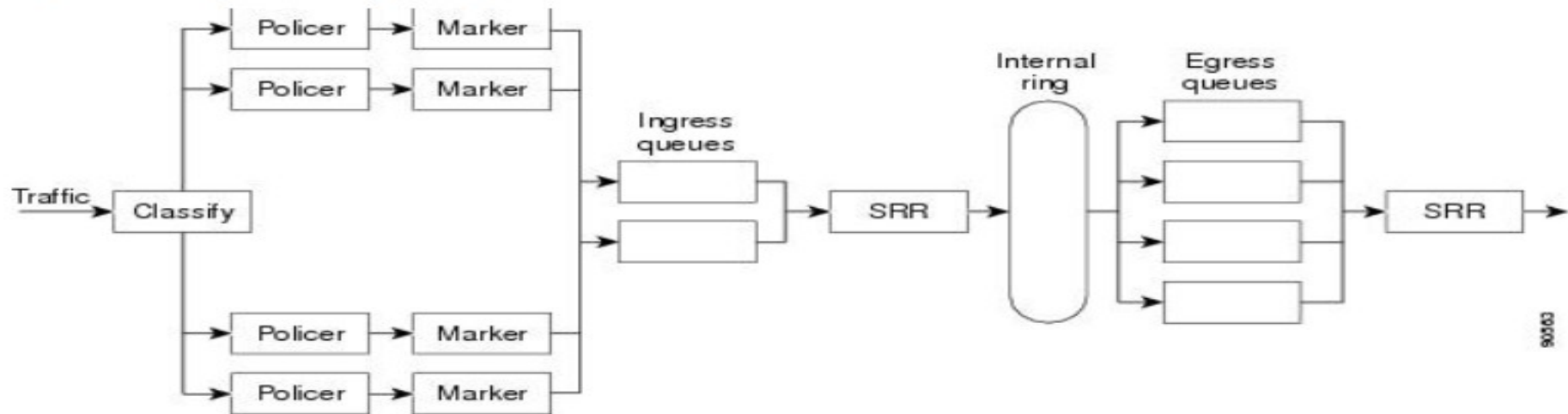
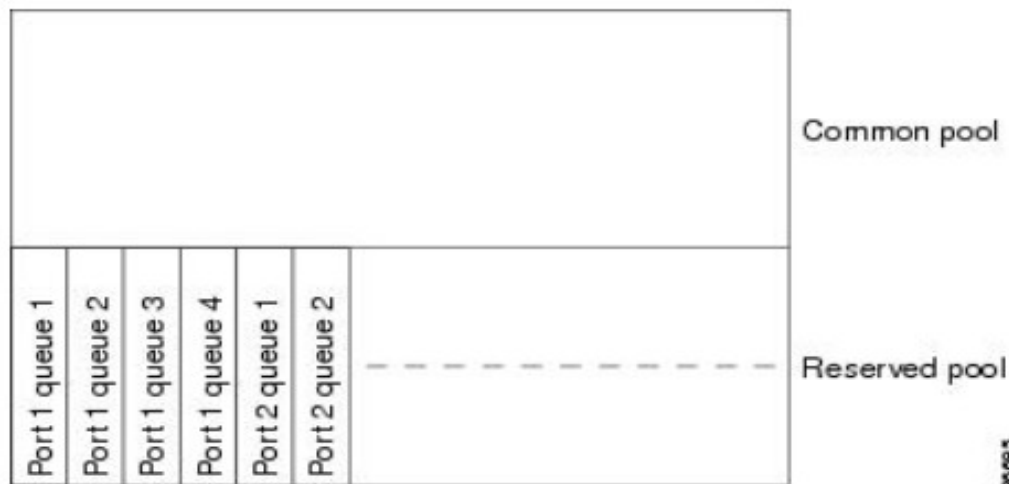


Figure 33-9 Egress Queue Buffer Allocation



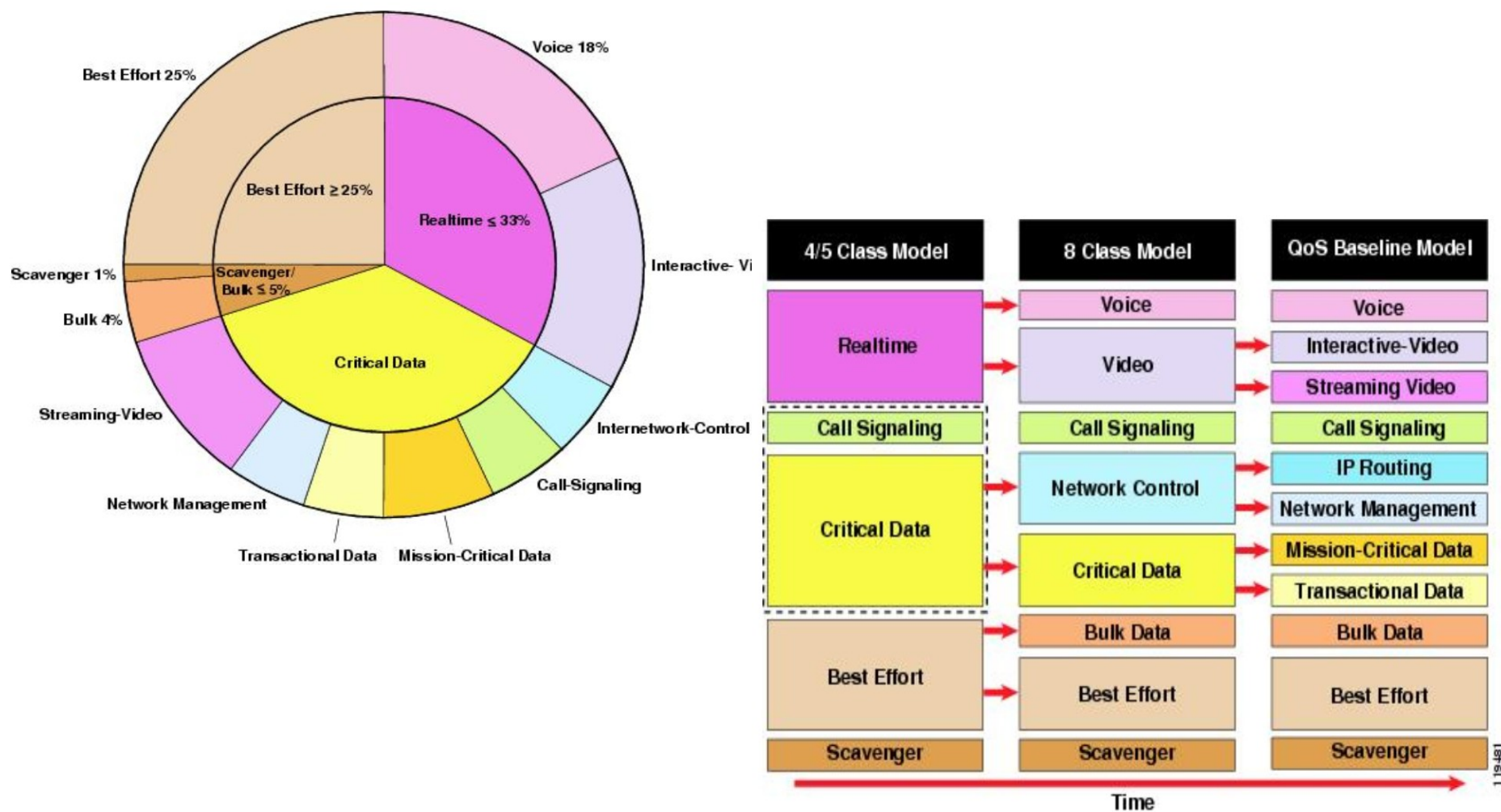
Traffic Flow Direction

Ingress != Egress

Egress Queue Quantity
→ Per Port → Maximum 4

Cisco - QoS Model (Defaults)

Figure 1-8 Compatible Four-Class and Eleven-Class Queuing Models following Realtime, Best Effort and Scavenger Queuing Rules



Cisco – QoS Label → Mappings

Table 33-2 Traffic Types, Packet Labels, and Queues

	VoIP ¹ Data Traffic	VoIP Control Traffic	Routing Protocol Traffic	STP BPDU Traffic	Real-Time Video Traffic	All Other Traffic
DSCP	46	24, 26	48	56	34	-
CoS	5	3	6	7	3	-
CoS-to-Ingress Queue Map	4, 5 (queue 2)					0, 1, 2, 3, 6, 7(queue 1)
CoS-to-Egress Queue Map	4, 5 (queue 1)	2, 3, 6, 7 (queue 2)			0 (queue 3)	2 (queue 3) 0, 1 (queue 4)

¹ VoIP = voice over IP

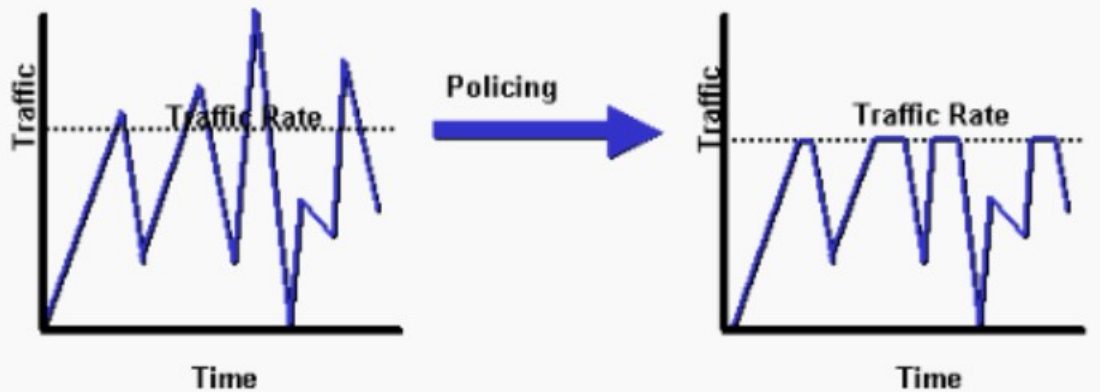
Table 33-3 Auto-QoS Configuration for the Ingress Queues

Ingress Queue	Queue Number	CoS-to-Queue Map	Queue Weight (Bandwidth)	Queue (Buffer) Size
SRR shared	1	0, 1, 2, 3, 6, 7	70 percent	90 percent
Priority	2	4, 5	30 percent	10 percent

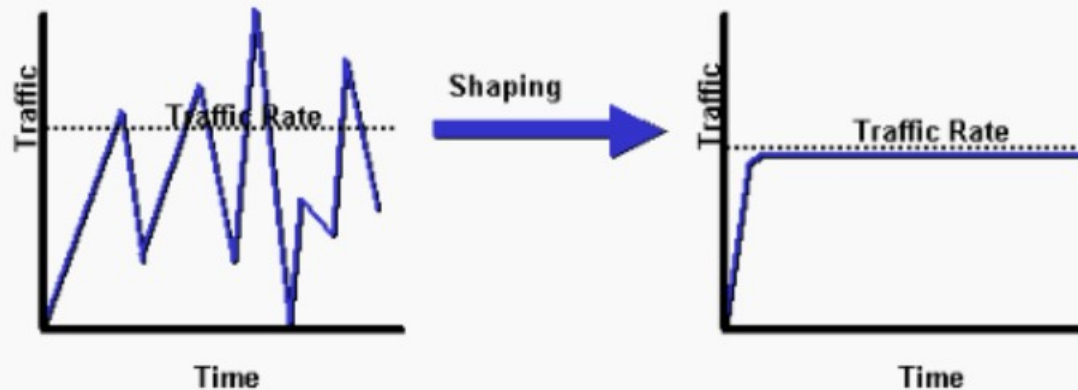
Table 33-4 Auto-QoS Configuration for the Egress Queues

Egress Queue	Queue Number	CoS-to-Queue Map	Queue Weight (Bandwidth)	Queue (Buffer) Size for Gigabit-Capable Ports	Queue (Buffer) Size for 10/100 Ethernet Ports
Priority	1	4, 5	up to 100 percent	25 percent	15 percent
SRR shared	2	2, 3, 6, 7	10 percent	25 percent	25 percent
SRR shared	3	0	60 percent	25 percent	40 percent
SRR shared	4	1	20 percent	25 percent	20 percent

Cisco - Traffic Policing vs Shapping for Bandwidth Control



In/Out-Bound
Max-Limits
Excess Drop
Excess Re-Mark (DSCP/IPP)
Lower Priority



Outbound
Memory
Buffers
Queues
Scheduling
→ Based Weighted Fair Queuing (CBWFQ)
→ Low Latency Queuing (LLQ).
Later Transmission

Cisco - Traffic Policing vs Shapping

	Shaping	Policing
Objective	Buffer and queue excess packets above the committed rates.	Drop (or remark) excess packets above the committed rates. Does not buffer.*
Token Refresh Rate	Incremented at the start of a time interval. (Minimum number of intervals is required.)	Continuous based on formula: 1 / committed information rate
Token Values	Configured in bits per second.	Configured in bytes.
Configuration Options	• shape command in the modular quality of service command-line interface (MQC) to implement class-based shaping. • frame-relay traffic-shape command to implement Frame Relay Traffic Shaping (FRTS). • traffic-shape command to implement Generic Traffic Shaping (GTS).	• police command in the MQC to implement class-based policing. • rate-limit command to implement committed access rate (CAR).
Applicable on Inbound	No	Yes
Applicable on Outbound	Yes	Yes
Bursts	Controls bursts by smoothing the output rate over at least eight time intervals. Uses a leaky bucket to delay traffic, which achieves a smoothing effect.	Propagates bursts. Does no smoothing.
Advantages	Less likely to drop excess packets since excess packets are buffered. (Buffers packets up to the length of the queue. Drops may occur if excess traffic is sustained at high rates.) Typically avoids retransmissions due to dropped packets.	Controls the output rate through packet drops. Avoids delays due to queuing.
Disadvantages	Can introduce delay due to queuing, particularly deep queues.	Drops excess packets (when configured), throttling TCP window sizes and reducing the overall output rate of affected traffic streams. Overly aggressive burst sizes may lead to excess packet drops and throttle the overall output rate, particularly with TCP-based flows.
Optional Packet Remarking	No	Yes (with legacy CAR feature).

Policing

rate-limit

- on interface → ACL
- ACL := MAC, IPP, MPLS EXP
- ACL → kbps

policy-map-class (bandwidth)

- on interface → POLICY
- POLICY → CLASS
- CLASS → kbps XOR %port

Shaping

policy-map-class (shape)

- average
- peak
- cir (comm. info. rate)
- bc (comm. burst size)
- be (excess burst)

Warning

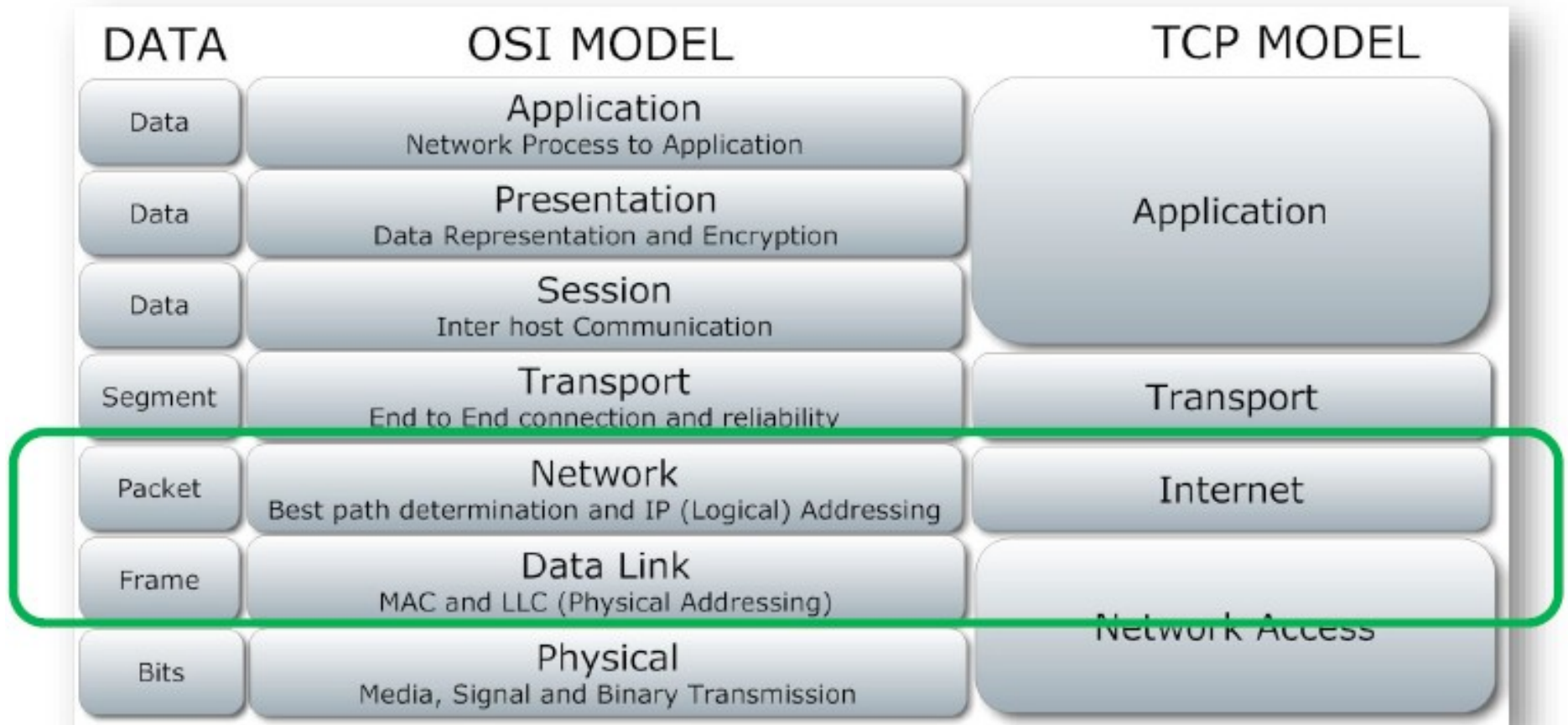
policy & priority => shape

- priority overrides police!!!

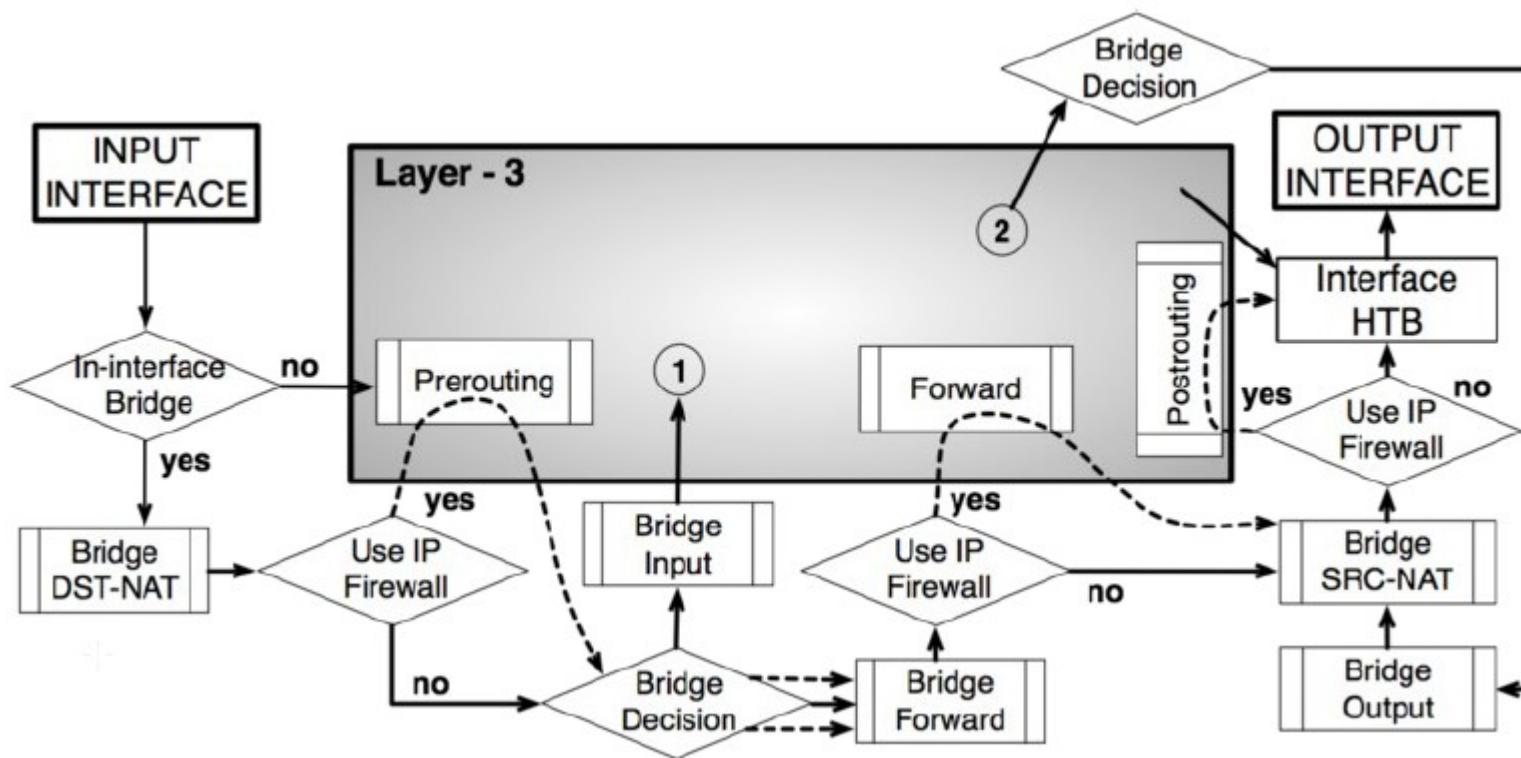
Mikrotik (RouteBand)

- Letonian Company
- Router OS Linux (Licensed, Never Expire)
 - <http://wiki.mikrotik.com/wiki/Manual:License>
- Layer 2 (Bridger) ↔ Layer 3 (Router)
- Hierarchical Token Bucket (Multi-Queue Tree)
- Disable Connection Tracking for all the router or by interfaces when is not needed NAT or Stateful Firewall to avoid asymmetric routed traffic drop (BGP).

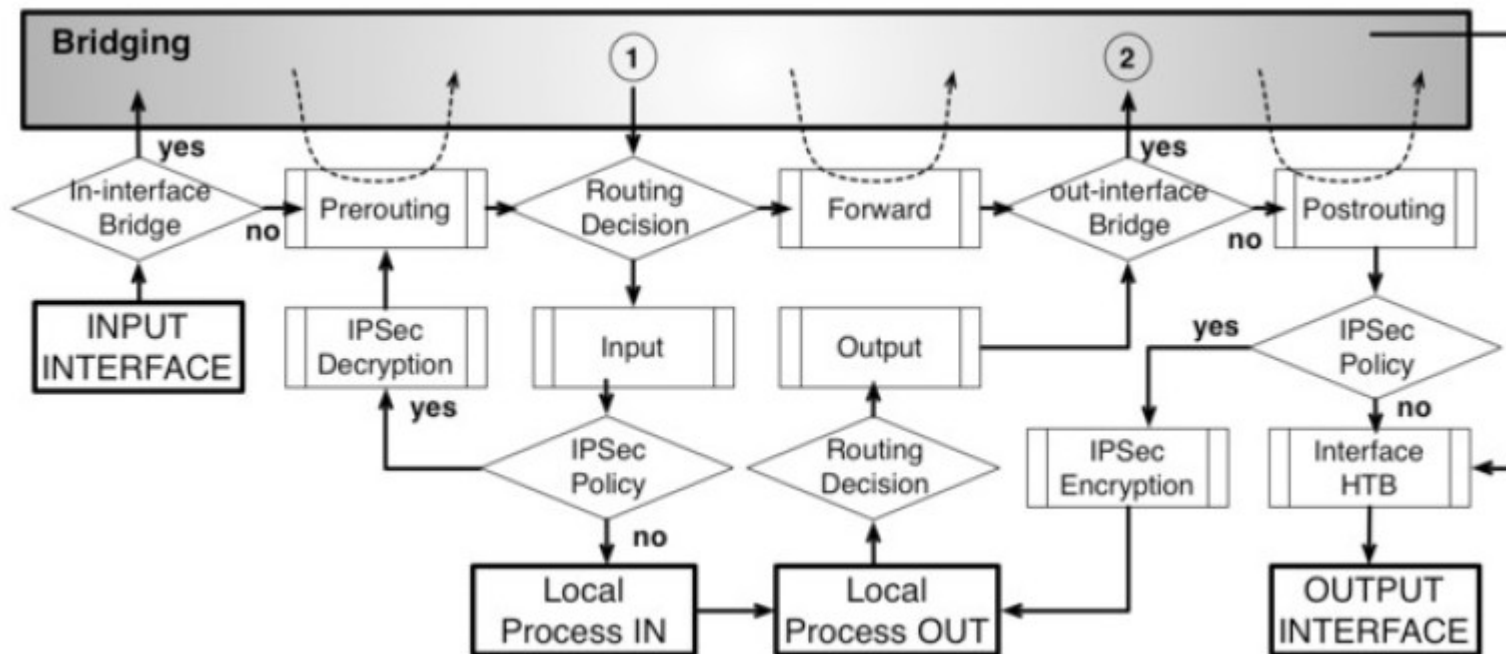
Mikrotik – Network Layers



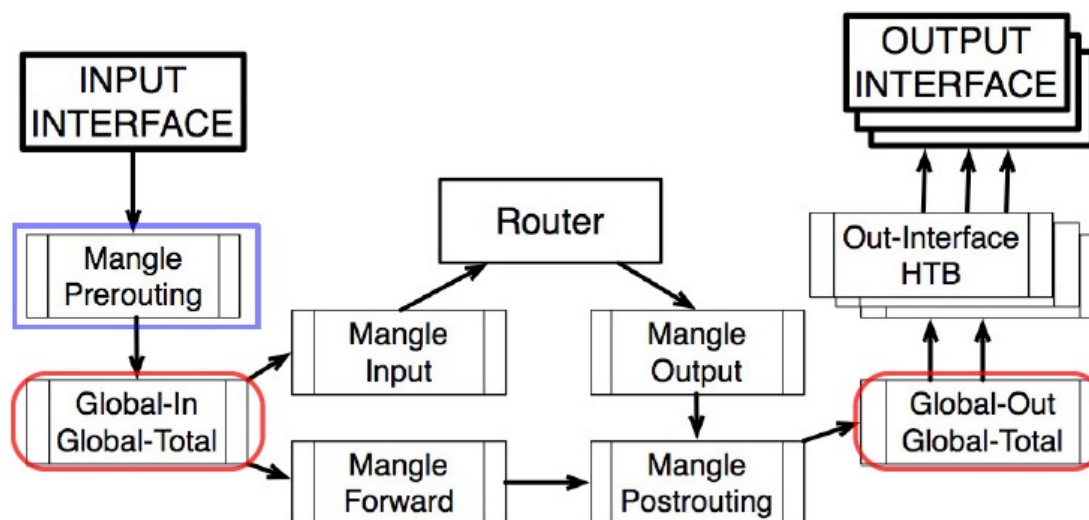
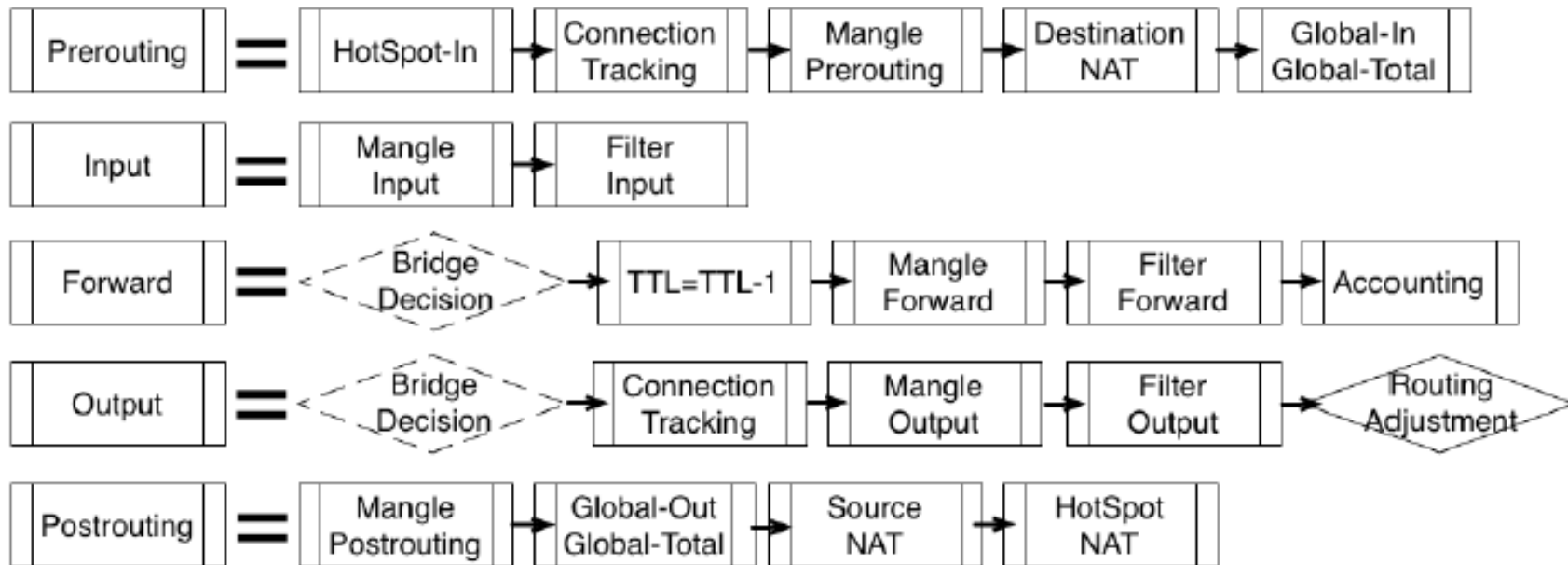
Mikrotik – Layer 2 - Bridging



Mikrotik – Layer 3 - Routing



Mikrotik – Traffic Flow

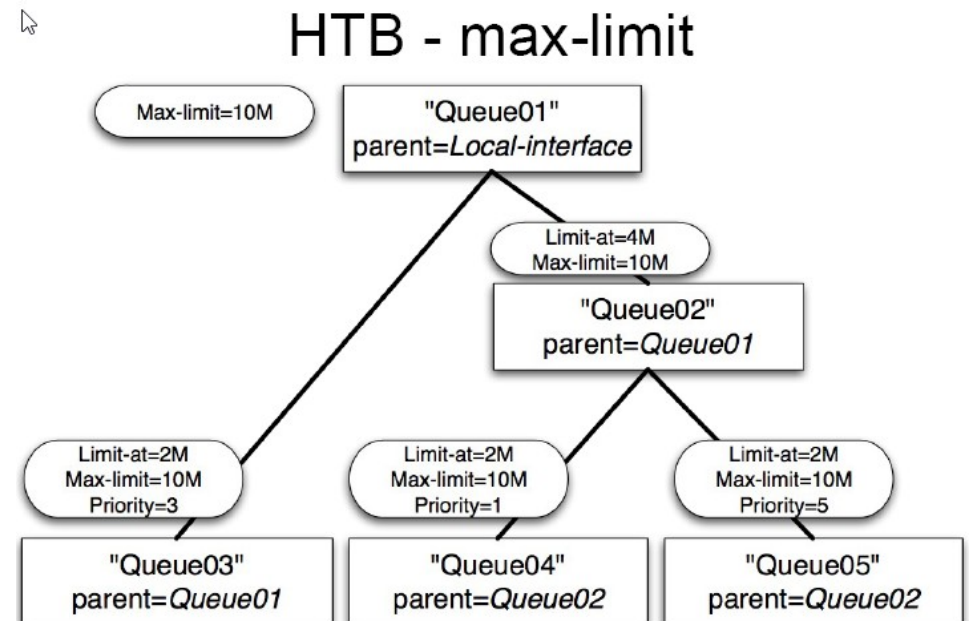
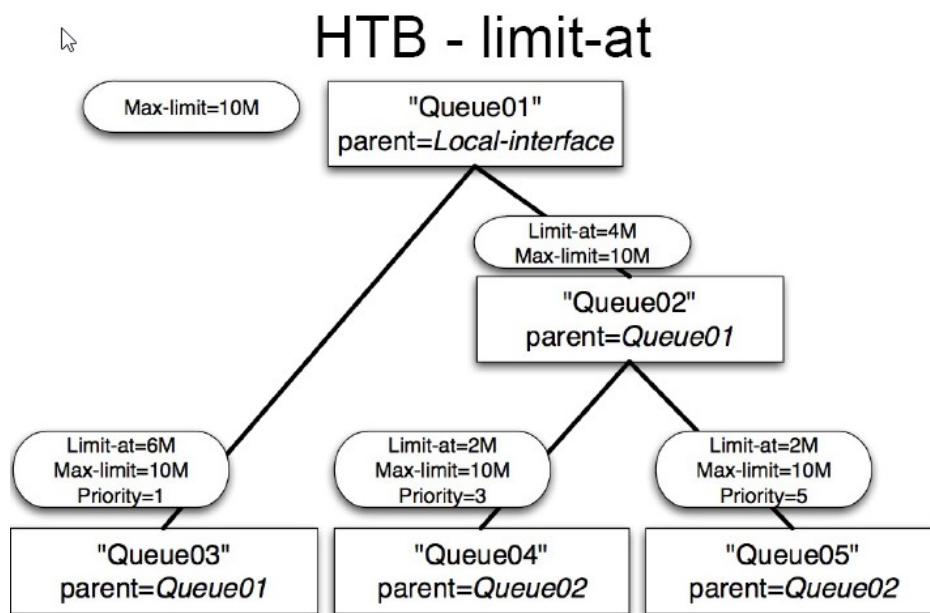


Mikrotik – Queueing

HTB Features – Dual Limitation

- HTB has two rate limits:
 - ◆ **CIR (Committed Information Rate)** – (limit-at in RouterOS) worst case scenario, flow will get this amount of traffic no matter what (assuming we can actually send so much data)
 - ◆ **MIR (Maximal Information Rate)** – (max-limit in RouterOS) best case scenario, rate that flow can get up to, if there queue's parent has spare bandwidth
- At first HTB will try to satisfy every child queue's **limit-at** – only then it will try to reach **max-limit**

Mikrotik – HTB Queue Example



Mikrotik – QoS Design

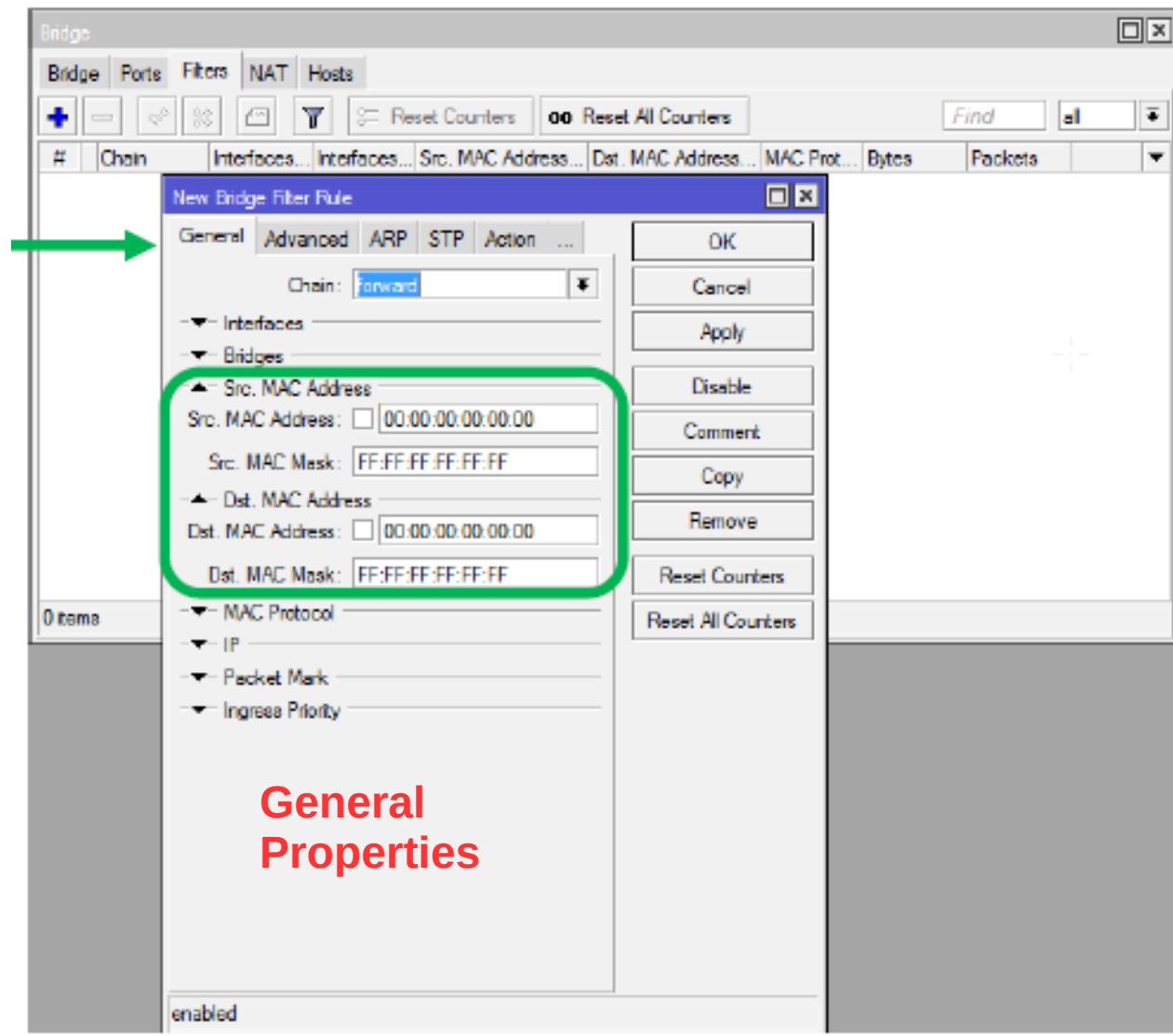
Layer 2 – Bridge Filter/Firewall

Mikrotik – Layer 2 – Ethernet Frame

Layer	Preamble	Start of frame delimiter	MAC Destination Address	MAC Source Address	802.1Q tag (optional)	Ethertype (Ethernet II) or length (IEEE 802.3)	Payload	Frame Check Sequence (32-bit CRC)	Interpacket Gap
	7 octets	1 octet	6 octets	6 octets	(4 octets)	2 octets	46(42 min with 802.1Q tag)–1500 octets	4 octets	12 octets
Layer 2 Ethernet frame			← 64–1518(1522) octets →						
Layer 1 Ethernet packet	← 72–1526(1530) octets →								

Portion which can be captured for analysis

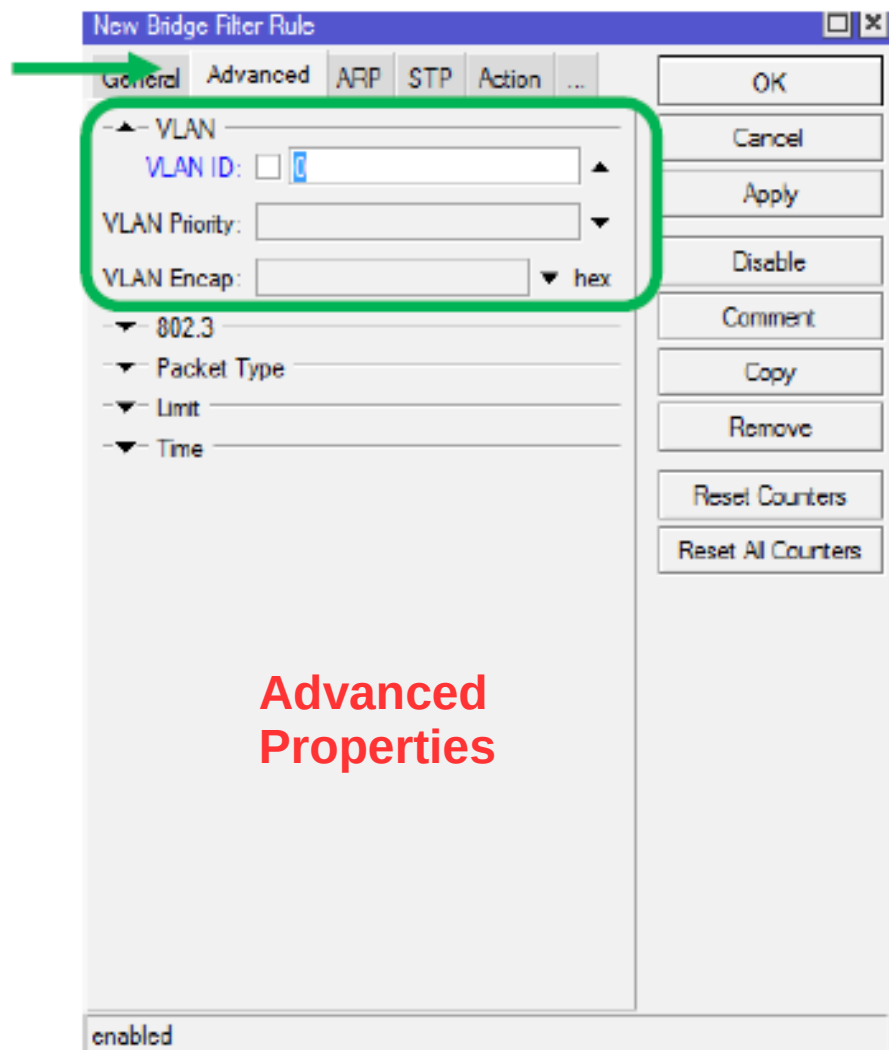
Mikrotik – Bridge Filter – Properties



Mikrotik – Bridge Filter - VLAN

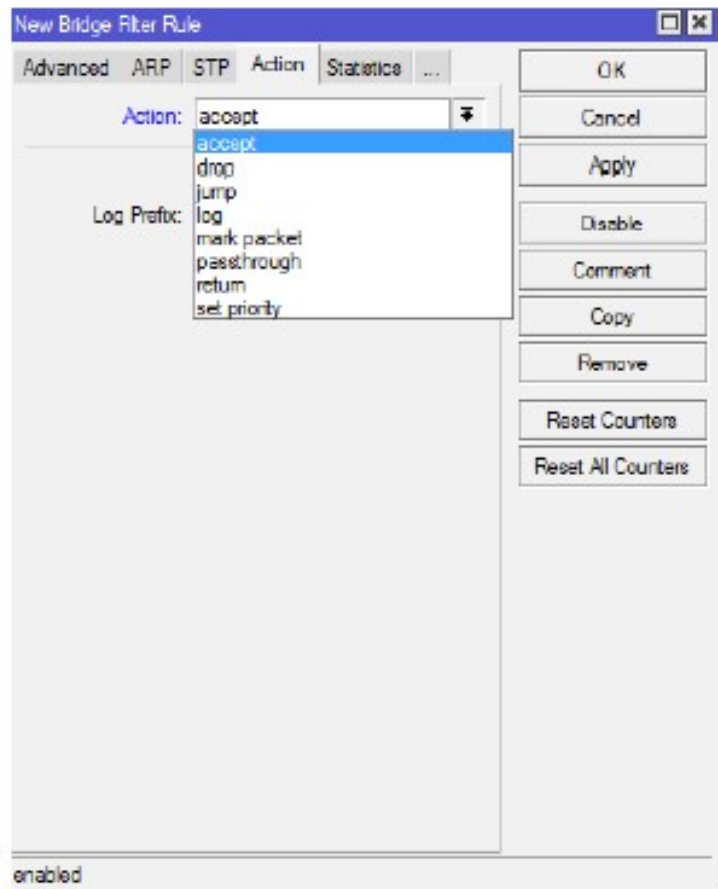
802.1Q Tag Field

- ▶ **vlan-id** (*integer 0..4095*)
- ▶ **vlan-priority** (*integer 0..7*)
- ▶ **vlan-encap** (*802.2 | arp | ip | ipv6 | ipx | length | mpls-multicast | mpls-unicast | ppoe | ppoe-discovery | rarp | vlan or integer: 0..65535 decimal format or 0x0000-0xffff hex format*)

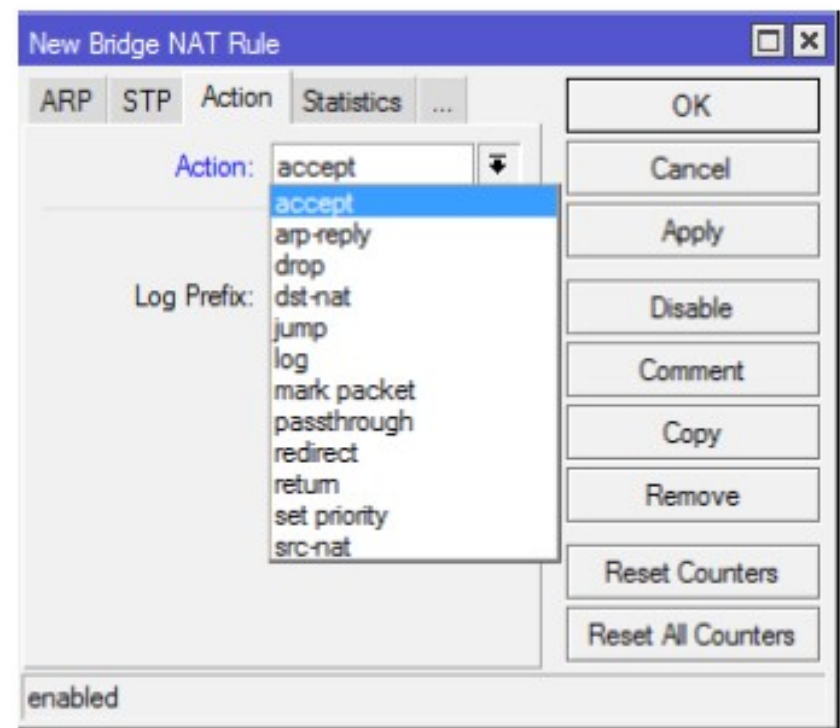


Mikrotik – Bridge Filter - Actions

Filter Actions



NAT Actions

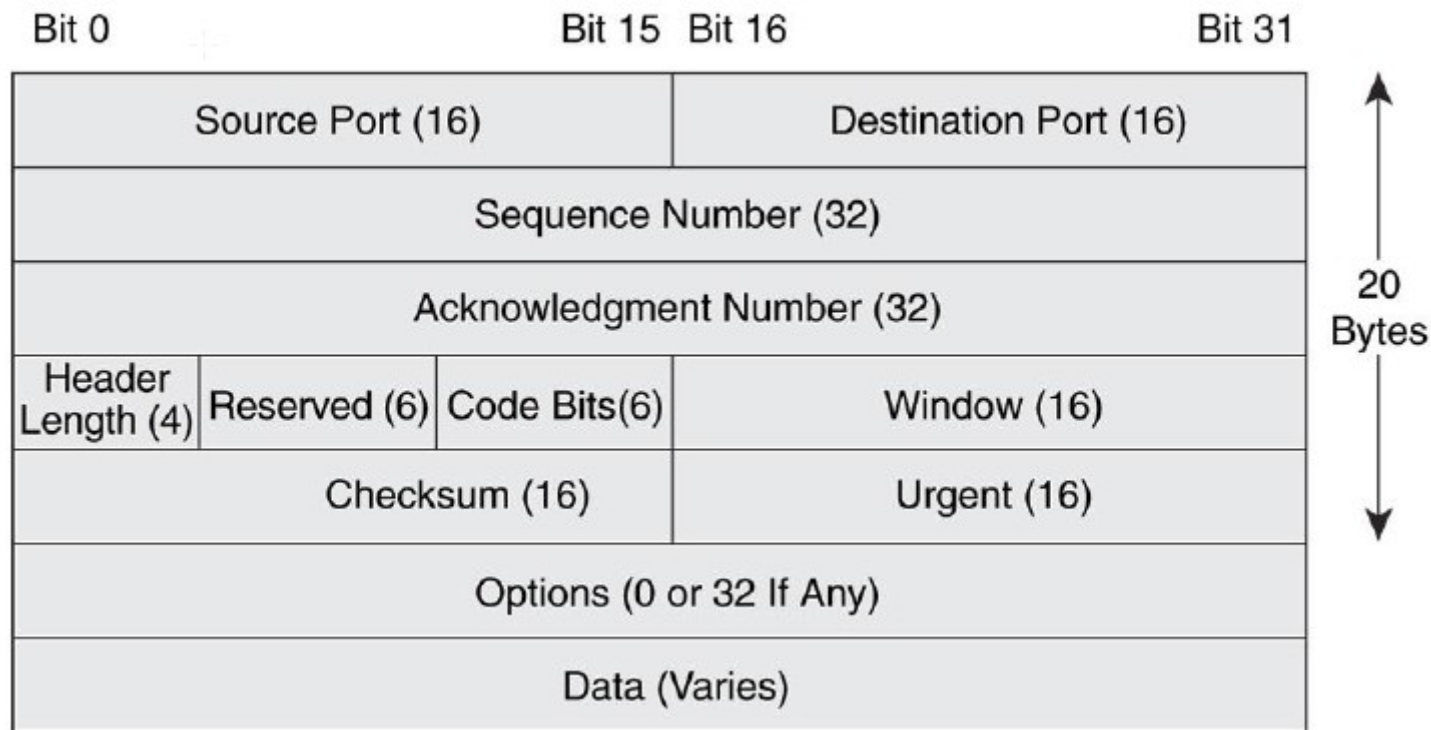


Mikrotik – QoS Design

Layer 3 – IP Filter / Firewall

Mikrotik – Layer 3 - Packet Header

L3 TCP Header (L2 Frame Payload)



Mikrotik – IP Filter - Properties

New Firewall Rule

General Advanced Action Statistics

Chain: forward

Src. Address:

Dst. Address:

Protocol: ☐ IP (tcp)

Src. Port: ☐

Dst. Port: ☐

Any Port: ☐

P2P:

In. Interface:

Out. Interface:

Packet Mark:

Connection Mark:

Routing Mark:

Routing Table:

Connection Type:

Connection State:

Connection NAT State:

enabled

OK Cancel Apply Disable Comment Copy Remove Reset Counters Reset All Counters

General Properties

Firewall Rule

General Advanced Action Statistics

Src. Address List:

Dst. Address List:

Layer7 Protocol:

Content:

Connection Bytes:

Connection Rate:

Per Connection Classifier:

Src. MAC Address:

Out. Bridge Port:

In. Bridge Port:

Ingress Priority:

Priority:

DSCP (TOS):

TCP MSS:

Packet Size:

Random:

TCP Rags: ☐ fin

ICMP Options:

IPv4 Options:

TTL:

OK Cancel Apply Disable Comment Copy Remove Reset Counters Reset All Counters

Advanced Properties

Mikrotik – IP Filter - Actions

Filters

Filter Rule

General Advanced Extra Action Statistics

Action: **drop**

Log Prefix:

OK Cancel Apply Disable Comment Copy Remove Reset Counters Reset All Counters

enabled

Nat

New NAT Rule

General Advanced Extra Action Statistics

Action: **src-nat**

Log Prefix:

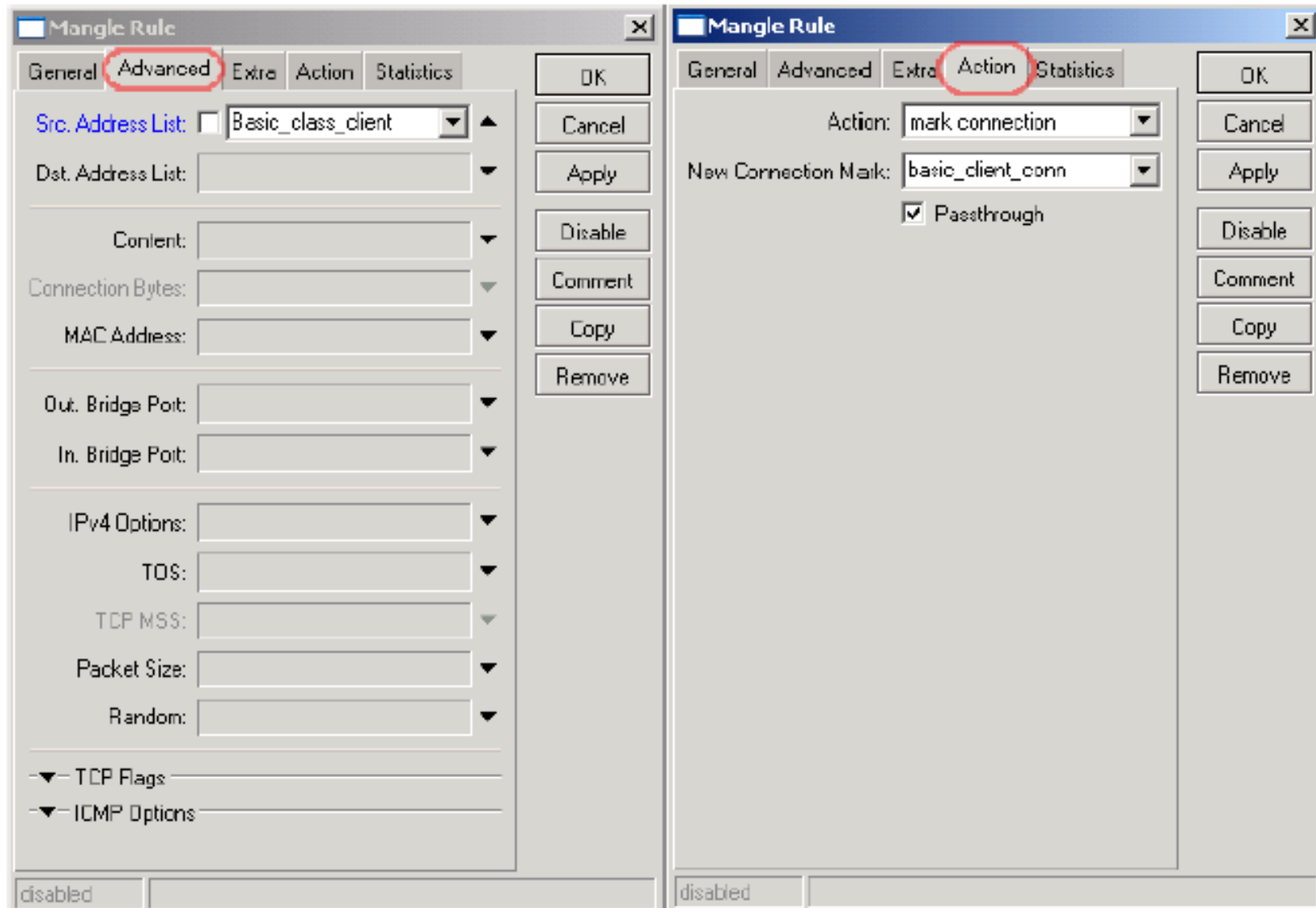
OK Cancel Apply Disable Comment Copy Remove Reset Counters Reset All Counters

enabled

Mikrotik – QoS Design

- **Traffic Queue Engineering:**
 - Parent Queue → Max-At: 100 Mbps
 - Children Queues → Limit-At: $X < 100$ Mbps
- **Bandwidth Control L2/L3:**
 - In → Interface (Iface) VLAN ID (~ any physical)
 - Bridge Filter (Mark DSCP → Iface VLAN ID)
 - Route Filter
 - Iface VLAN ID & Mark DSCP & Address (Src): 0.0.0.0
 - Connection-Mark → Child Queue #1

Mikrotik – QoS Design



Mikrotik – QoS Design

Firewall

Filter Rules NAT Mangle Service Ports Connections Address Lists

00 Reset Counters 00 Reset All Counters forward

#	Action	Chain	New Packet Mark	New Connection Mark	Bytes	Packets
::: mark basic client traffic						
	mark connection	forward		basic_client_conn	9893.1 MiB	18 599 504
	mark packet	forward	basic_client_traffic		22575.4 MiB	35 292 323
::: mark standard client traffic						
	mark connection	forward		standard_client_conn	825.4 MiB	2 747 515
	mark packet	forward	standard_client_traffic		6396.7 MiB	7 349 895
::: mark bussiness client traffic						
	mark connection	forward				
	mark packet	forward		busine		
::: Check for unmarked traffic						
	log	forward				

Queue List

Simple Queues Interface Queues Queue Tree Queue Types

00 Reset Counters 00 Reset All Counters

Name	Parent	Packet Mark	Limit At	Max Limit
Total_download	local_ether1		0	0
basic_client_download	Total_download	basic_client_traffic	0	0
business_client_download	Total_download	business_client_traffic	0	0
standard_client_download	Total_download	standard_client_traffic	0	0
Total_upload	public_ether3		0	0
basic_client_upload	Total_upload	basic_client_traffic	0	0
business_client_upload	Total_upload	business_client_traffic	0	0
standard_client_upload	Total_upload	standard_client_traffic	0	0

0 B queued 0 packets queued

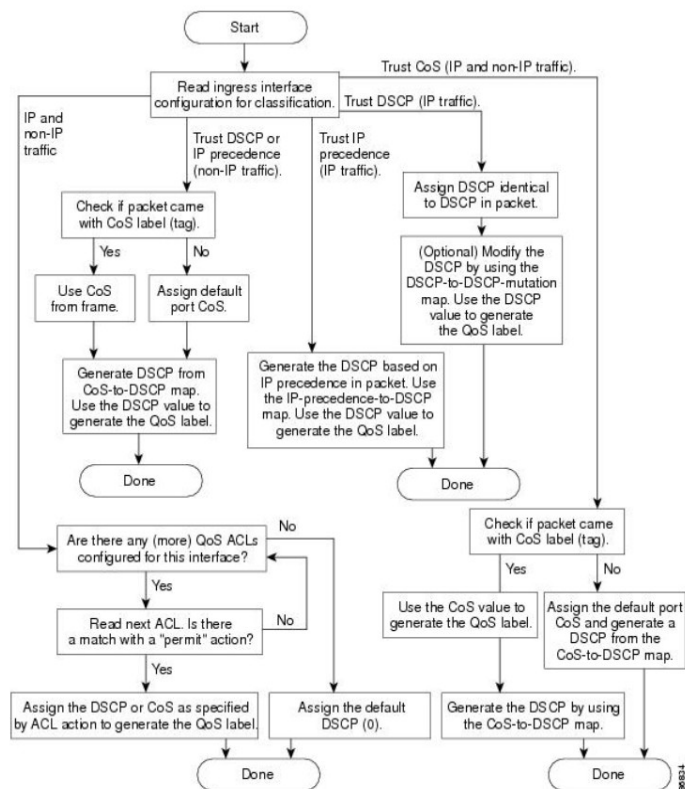
Mikrotik – Alternatives

- NetEqualizer → NetEqualizer 1000/5000
- Exinda → Exinda 4000/6000/8000/10000
- Blue Coat → Packet Shaper 3500/7500/12000
- Allot Communications → Service Gateway
Tera/9500/Virtual Edition/NetXplorer

APPENDIX A. Cisco QoS Extras

QoS Flows

Figure 33-3 Classification Flowchart



Queueing and Scheduling Overview

The switch has queues at specific points to help prevent congestion as shown in [Figure 33-5](#).

Figure 33-5 Ingress and Egress Queue Location

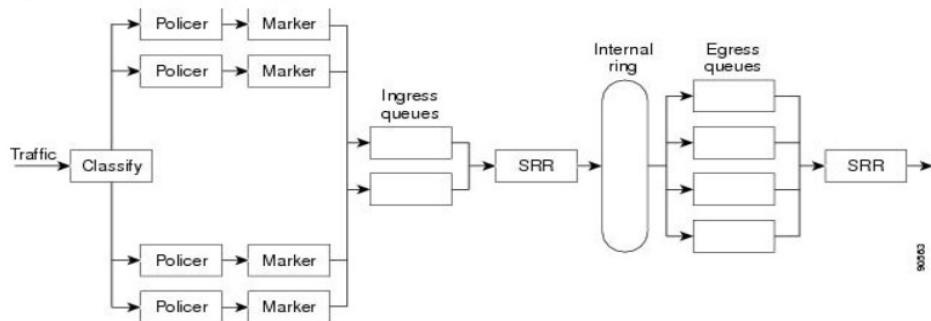


Figure 33-4 Policing and Marking Flowchart on Physical Ports

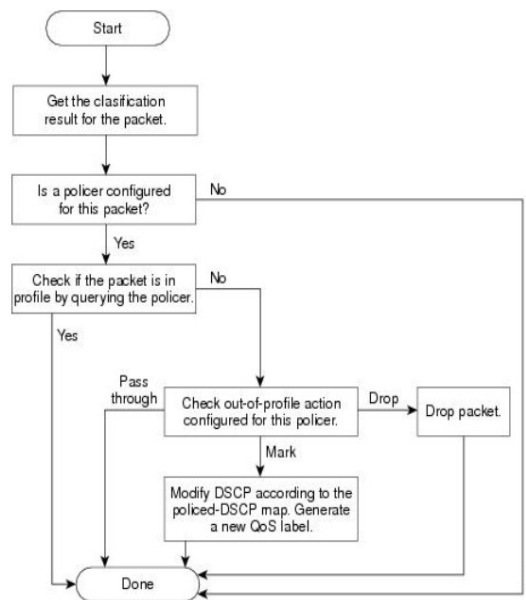


Figure 33-8 Queueing and Scheduling Flowchart for Egress F

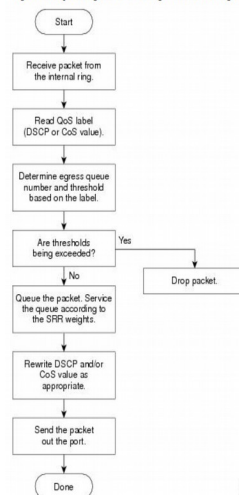


Figure 33-6 WTD and Queue Operation

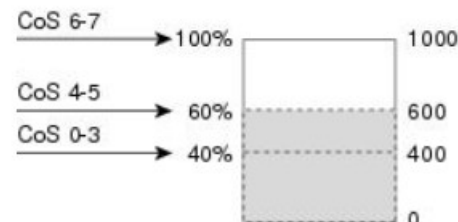


Figure 33-7 Queueing and Scheduling Flowchart for Ingress Ports

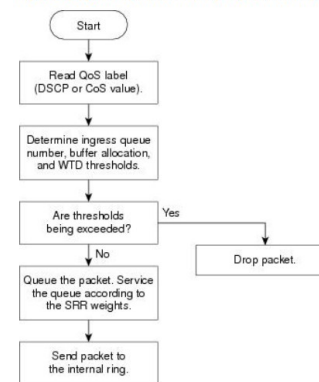
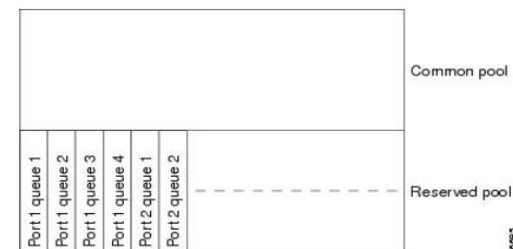
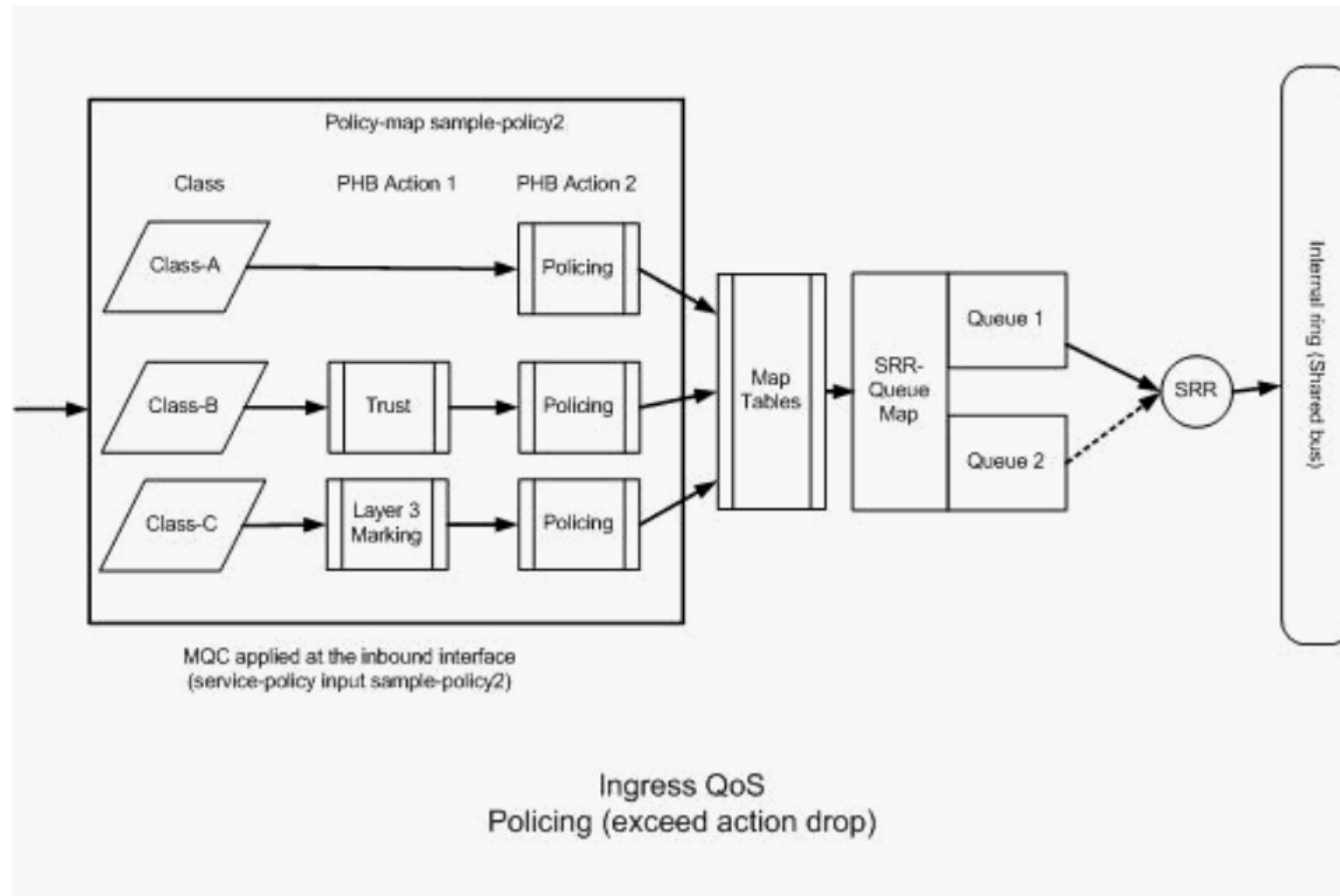


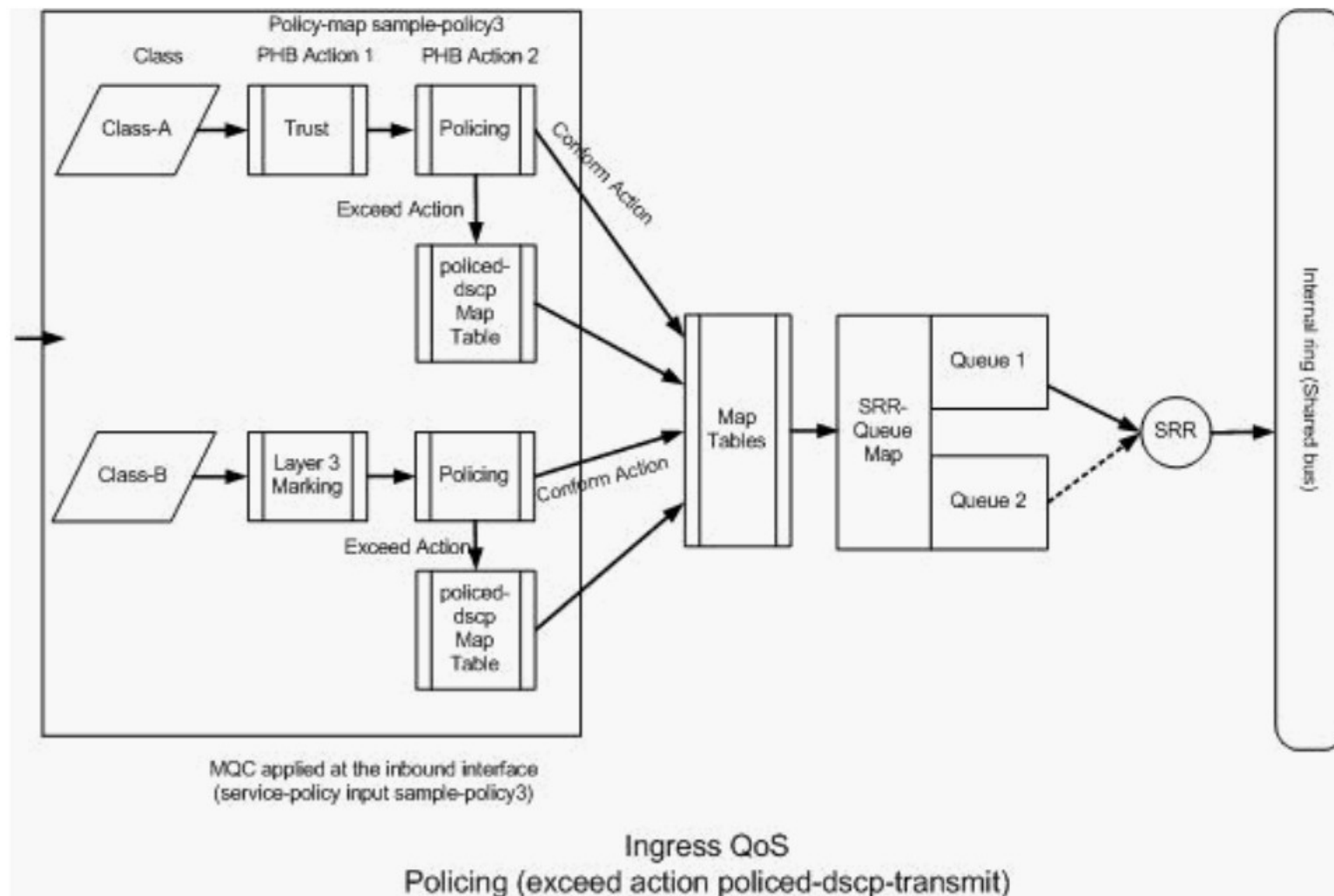
Figure 33-9 Egress Queue Buffer Allocation



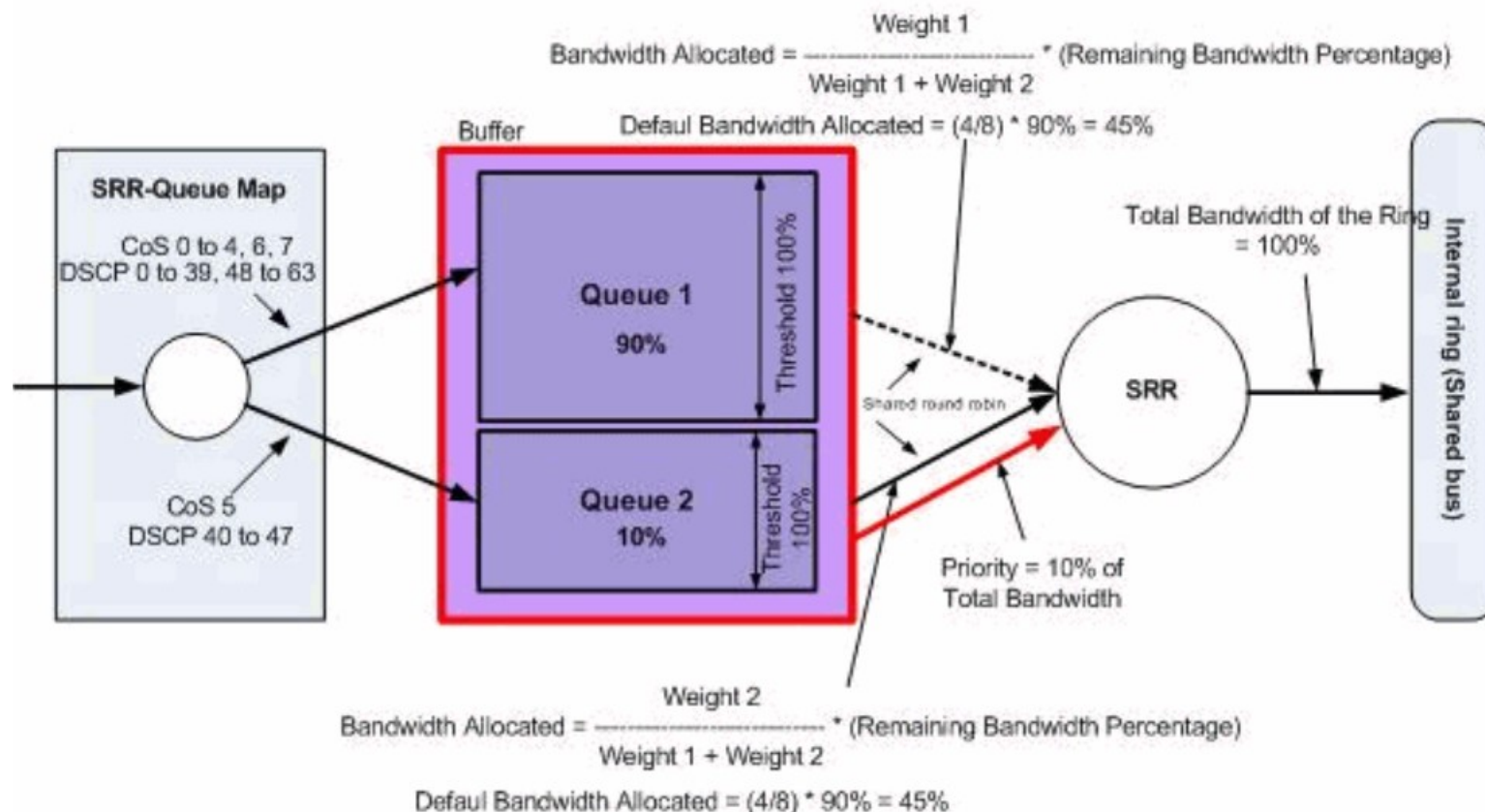
Clasif./Mark./Policing - Ingress QoS



Clasif./Mark./Policing - Ingress QoS

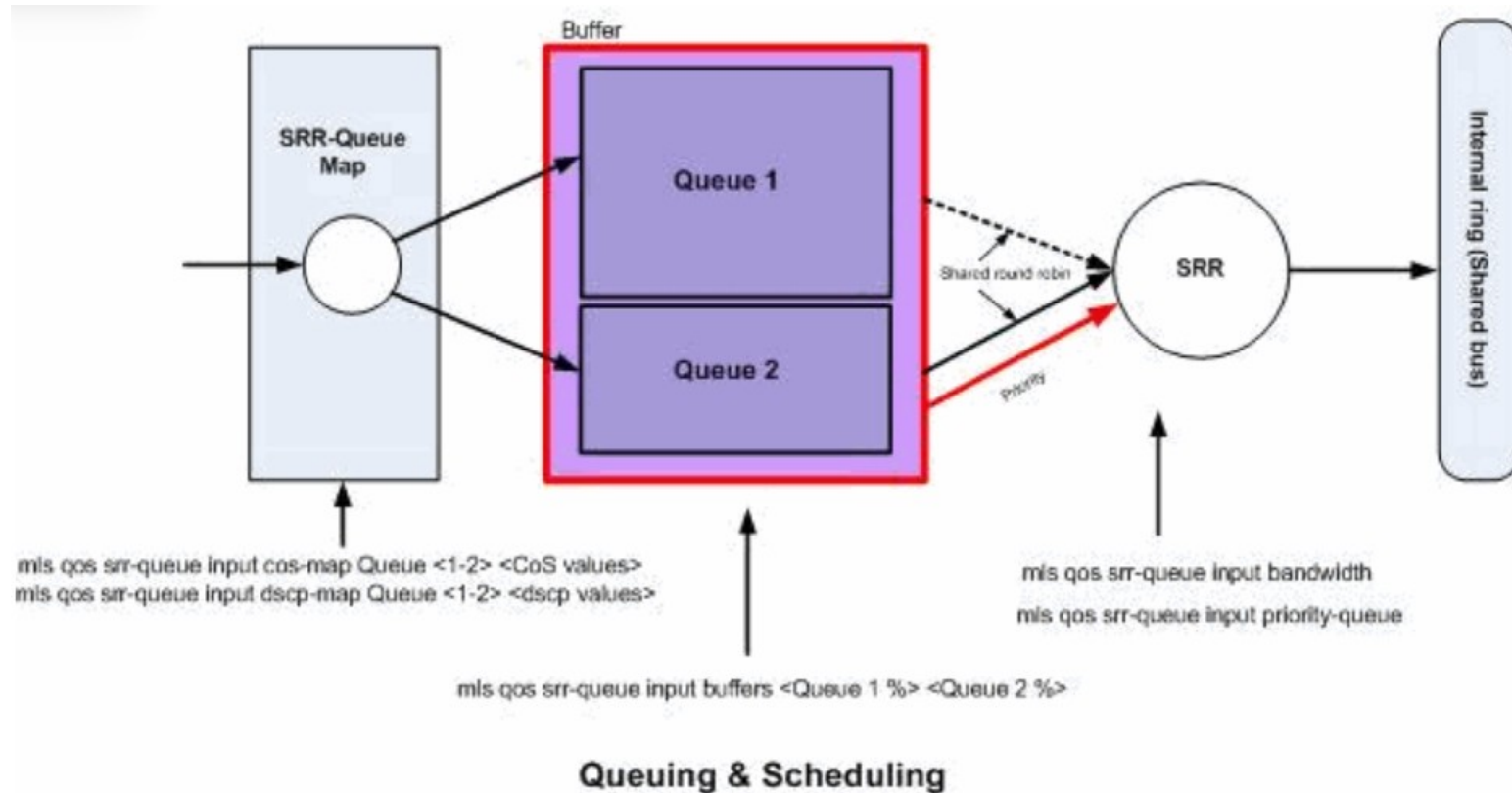


Congestion Management

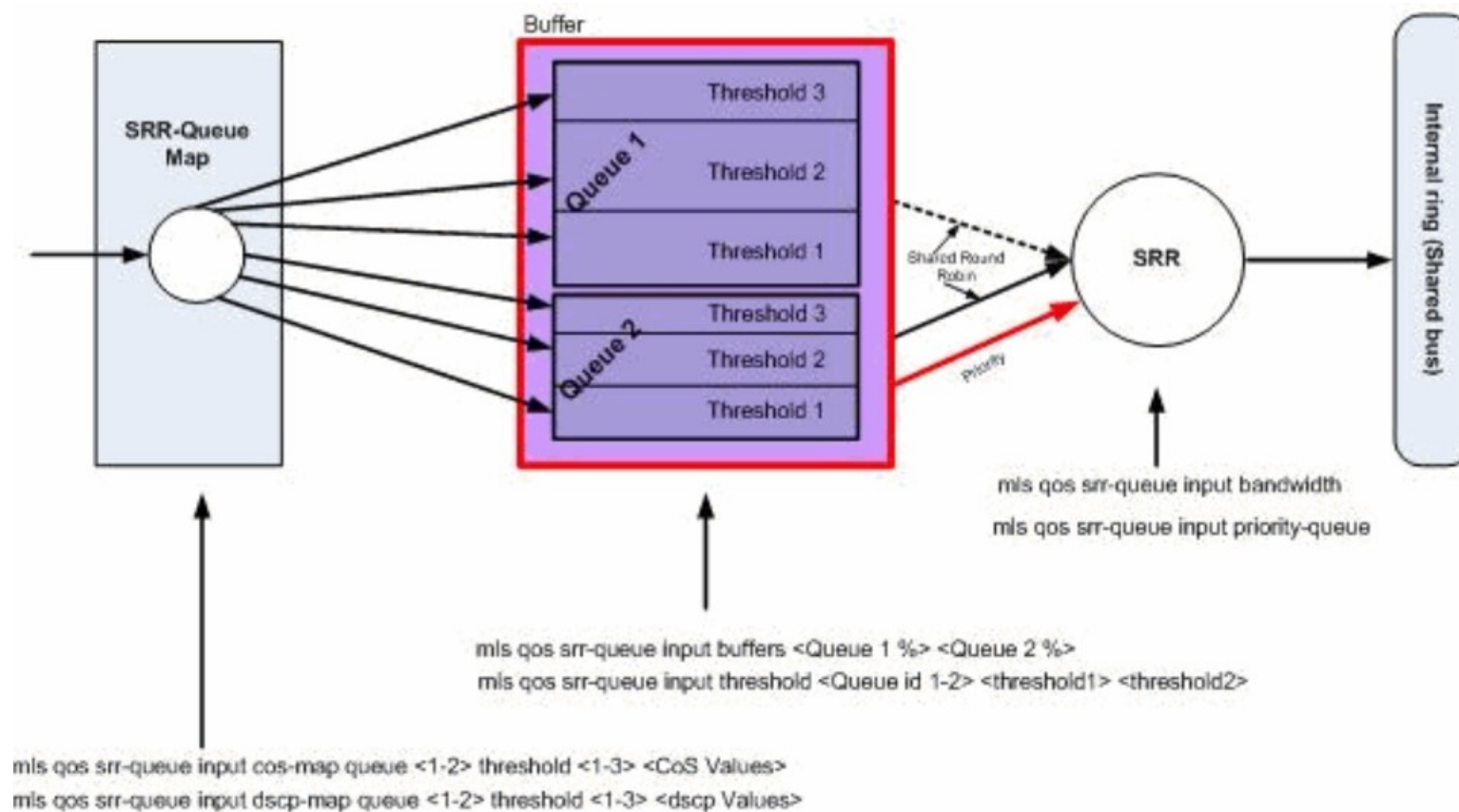


Default Queuing, Dropping & Scheduling

Congestion Management



Congestion Management

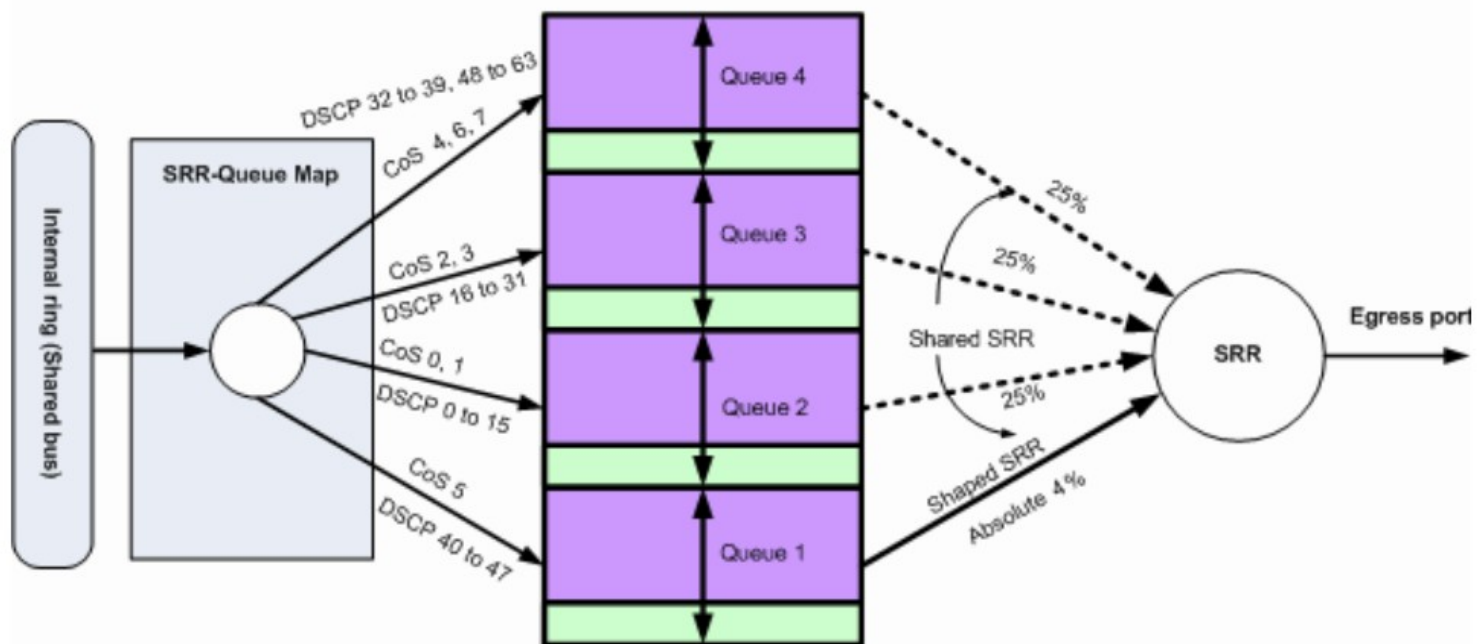


Queuing, Dropping & Scheduling

Congestion Management

Default Configuration

Default Queue Map Configuration:



Default Queueing, Dropping & Scheduling